

Telecom Portfolio Risk Report

Generated 12 August 2026 · portfolio state, not a time series

Unmarked	OBSERVED - measured from the source data.
†	PROJECTED - derived from the stated economic assumptions, not measured.
‡	SIMULATED - from generated history. Shows portfolio structure only; churn in it is flat by construction and cannot support a forecast.

Portfolio at a glance

3,333	14.49%	483
Customers	Churn rate	Churned
\$198,146	\$59.45	\$31,567
Revenue	ARPU	Revenue at risk

The most decision-relevant fact is that 14.49% of customers have churned, resulting in \$31,566.93 of revenue at risk. The confirmed churn drivers are 4+ customer service calls, international plan subscription, and heavy daytime usage, defined as a day charge of 45 or more. Customers with international plan subscriptions, such as the 323 subscribers, need attention first as they are one of the key drivers of churn. The total revenue of \$198,146.03 is also at risk due to these churn drivers. With 483 customers already churned, it is essential to focus on the high-risk groups to mitigate further revenue loss. The average revenue per user of \$59.45 is also a key metric to consider when developing strategies to retain customers.

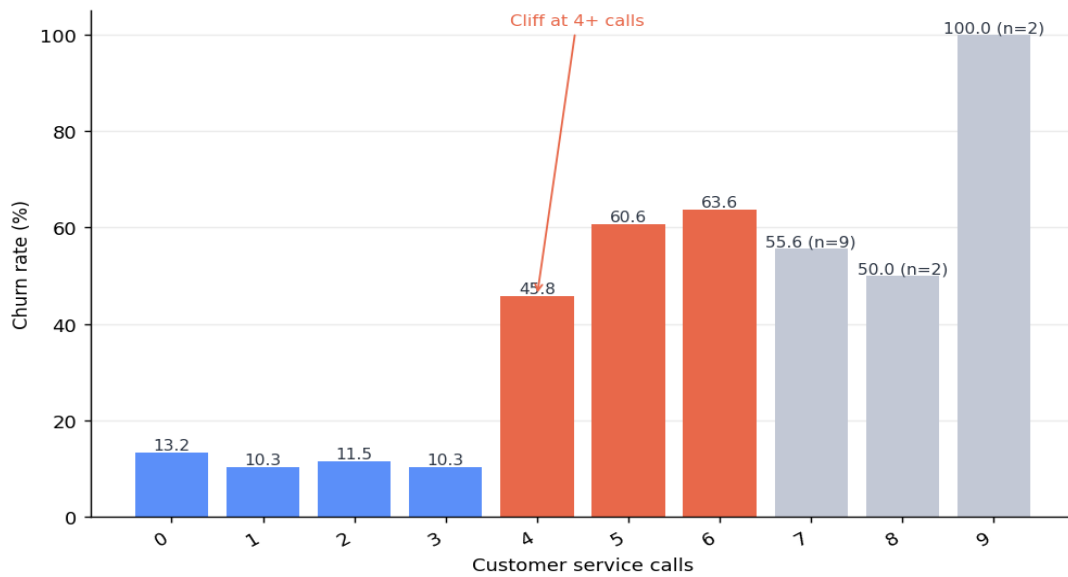
Confirmed churn drivers

We have identified three confirmed drivers of churn: international plan subscription, 4 or more customer service calls, and heavy daytime usage, defined as a day charge of 45 or more. Each of these drivers represents a threshold, rather than a gradual trend, which is important for targeting. For example, having 4 or more customer service calls raises the churn rate from 11.25% to 51.69%. Similarly, heavy daytime usage raises churn from 5.0% to 59.0% among customers with no other risk factors. International plan subscription also has a significant impact, with a churn rate of 43.72% for customers with this plan and no voicemail. These thresholds suggest that targeting customers who have crossed these thresholds can be an effective way to address churn.

Driver	Threshold	Below	Above	Customers
Customer service calls	4 or more	11.25%	51.69%	267
International plan	subscribed	11.5%	42.41%	323
Daytime charge	\$45 or more	11.43%	60.0%	210

Churn by number of customer service calls

Churn is flat from 0 to 3 calls, then jumps sharply. This is a cliff, not a slope.

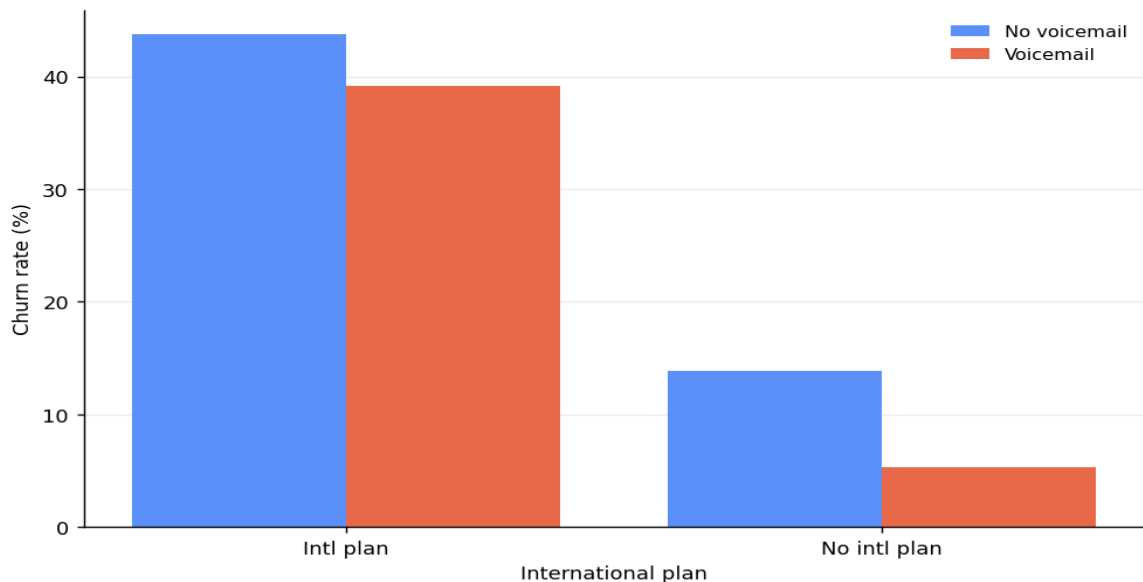


Bucket as '0-3' vs '4+'. Treating call count as a continuous predictor understates the effect badly. Greyed bars have fewer than 20 customers and are too small to read as evidence.

AI explanation. The pattern in the chart shows that the churn rate remains relatively flat at around 10-13% for customers with 0, 1, 2, and 3 service calls, but then jumps sharply to 45.78% at the fourth call. This suggests that a fourth call may imply that the first three calls did not resolve the issue, leading to increased frustration and a higher likelihood of churn. A possible explanation is that the customer may already have decided to leave before calling, and the fourth call is a sign of unresolved issues. To address this, the company should review and improve its issue resolution process for customers who have made multiple service calls, such as implementing a more effective escalation procedure to resolve issues before they reach the fourth call.

Churn by plan combination

International plan holders churn several times higher.

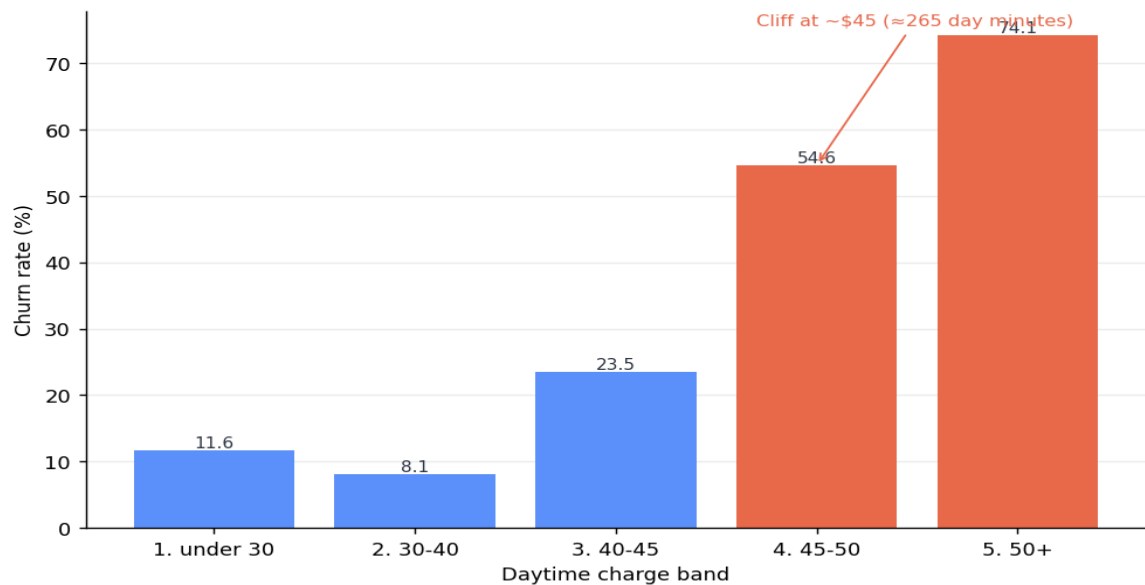


Voicemail plan has little independent effect; the international plan is the driver.

AI explanation. The pattern shows a sharp change in churn rate when the international plan is present, with rates of 43.72% and 39.13% for international plan holders, compared to 13.85% and 5.3% for those without the international plan. This suggests that the international plan may be a key factor in driving churn, possibly due to poor value for typical usage or higher price sensitivity among these customers. The voicemail plan, on the other hand, appears to have a minimal impact, with churn rates varying only slightly between voicemail and no voicemail groups. To address this issue, the company should review the pricing and benefits of the international plan to determine if adjustments can be made to improve its value proposition and reduce churn.

Churn by daytime charge band

Churn DIPS before it explodes. A linear model of day charge would find almost nothing.



Found while building the Phase 6 rules engine. The earlier two-driver segmentation reported a 'baseline' of 8.2% that concealed this group.

AI explanation. The pattern of churn rates shows a dip from 11.64% in the under 30 daytime charge band to 8.12% in the 30-40 band, before rising steeply to 23.53% in the 40-45 band and further to 54.61% and 74.14% in the higher bands. This non-monotonic pattern suggests that customers may be experiencing bill shock or are on the wrong plan for their usage, leading to high churn rates. A possible explanation is that heavy day callers, such as those with an average of 276.4 day minutes, may be on plans that do not suit their usage, prompting them to shop around. To address this, the company should review the plans of customers in the higher daytime charge bands, such as the 45-50 and 50+ bands, and consider offering more tailored plans to reduce churn.

Risk segments and recommended actions

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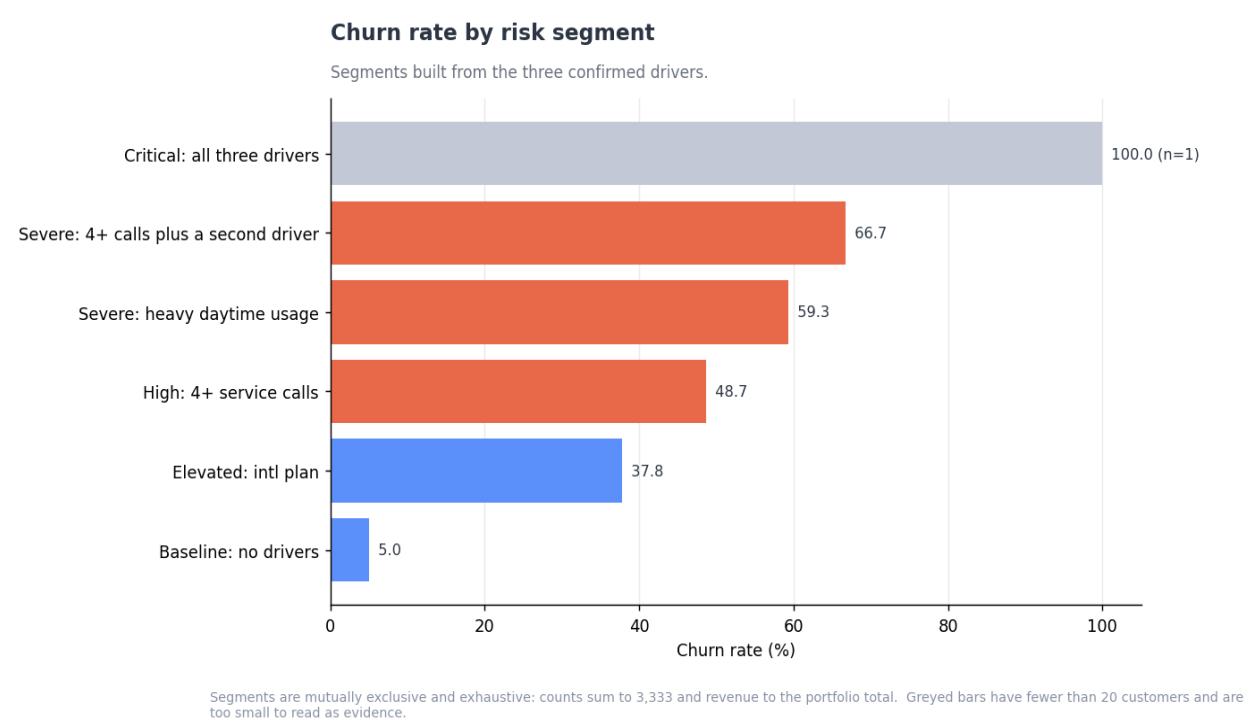
Segment	Customers	Churn	Revenue at risk	Net value†
Critical — all three risk drivers	1	100.0%	\$82	\$242†
Severe — 4+ service calls with a second driver	42	66.67%	\$1,834	\$5,752†
Severe — heavy daytime usage	194	59.28%	\$9,226	\$29,017†

Segment	Customers	Churn	Revenue at risk	Net value†
High — 4+ service calls only	224	48.66%	\$5,549	\$20,044†
Elevated — international plan only	267	37.83%	\$6,139	\$20,823†
Baseline — no risk drivers	2,605	4.95%	\$8,738	-

Recommended actions

Segment	Action	Offer	Customers
Critical — all three risk drivers	Immediate senior retention call, same day	Premium support + tariff review + 20% discount, 3 periods	1
Severe — 4+ service calls with a second driver	Senior retention specialist call within 48 hours	Premium support tier + 15% discount for 3 periods	42
Severe — heavy daytime usage	Tariff review call within 5 working days	Move to a daytime bundle or higher-allowance plan	194
High — 4+ service calls only	Proactive support review call within 5 working days	Service credit + dedicated support contact	224
Elevated — international plan only	Plan review — check the tariff fits actual usage	Tariff optimisation or downgrade without penalty	267

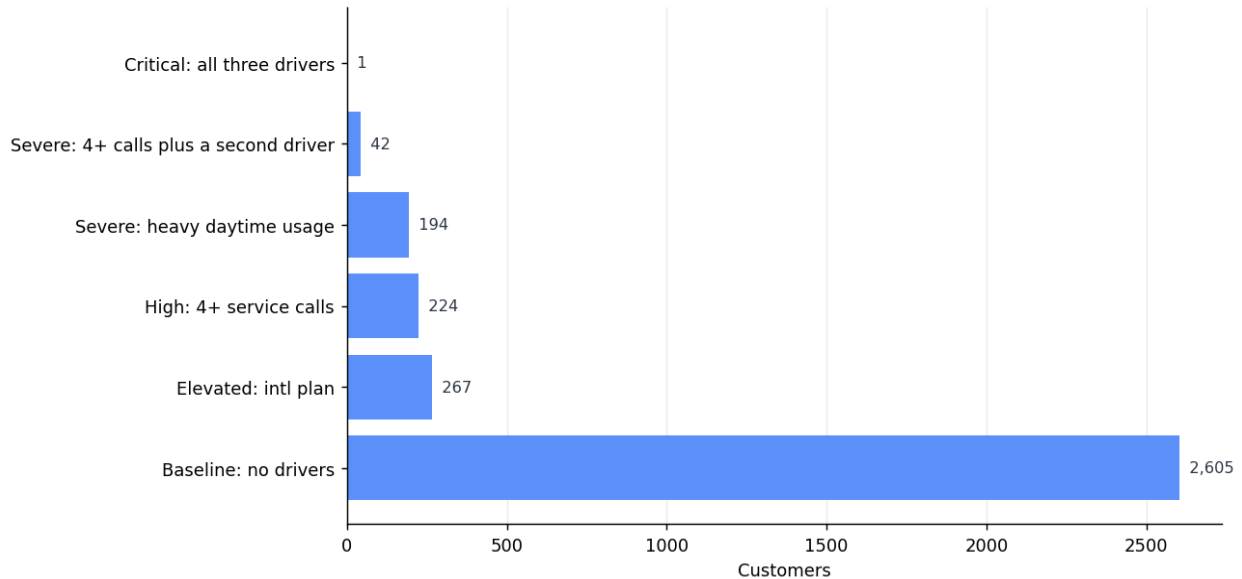
† **Assumptions behind projected values:** save rate 30% (band 15%-45%), contact cost \$8.00, acquisition cost \$45.00, horizon 12 periods. These are industry-typical placeholders, not measurements: the dataset contains no cost data.



AI explanation. The pattern shows a sharp change in churn rate at the "Severe: 4+ calls plus a second driver" and "Severe: heavy daytime usage" segments, with churn rates of 66.67% and 59.28%, respectively. The "Severe: heavy daytime usage" segment deserves attention first, as it has a large number of customers (194) and a significant revenue at risk (9225.97). This suggests that customers with heavy daytime usage may be experiencing issues with their service, leading to a higher churn rate. To address this, the company should review the service plans and usage patterns of this segment to identify potential areas for improvement, such as offering more tailored plans or improving network coverage.

Customers per risk segment

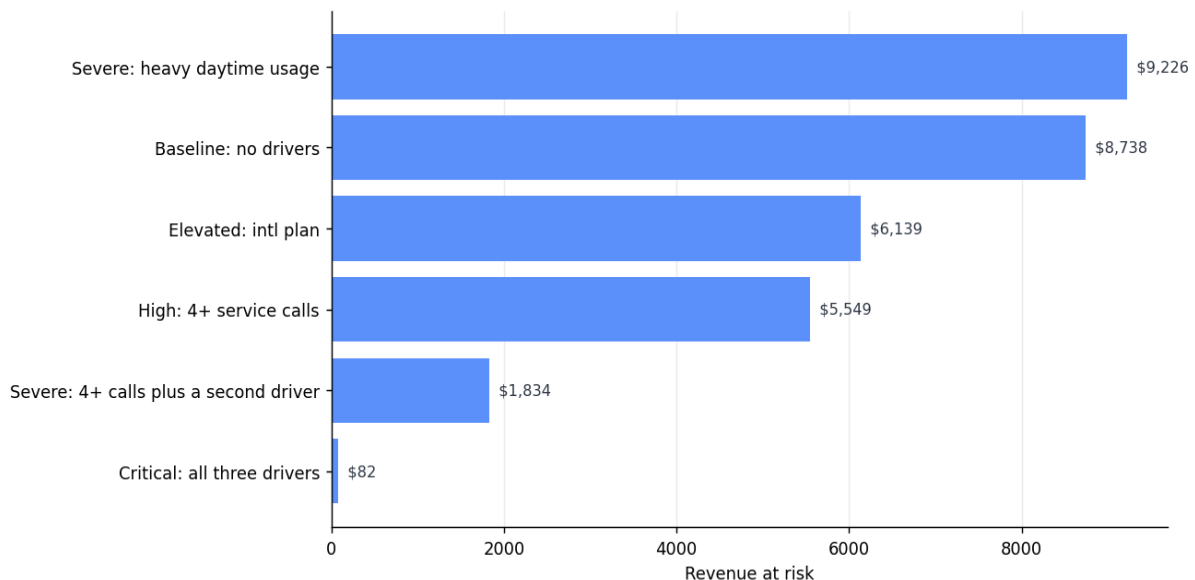
Most of the base carries no risk driver at all.



Segment sizes are highly uneven; read rates alongside n.

Revenue at risk by segment

Revenue attached to customers who churned, by segment. The largest exposure is not the highest-rate segment.



Rate and exposure rank differently. A large low-rate segment can carry more absolute revenue risk than a small high-rate one.

AI explanation. The pattern shows that the segment with the highest churn rate, "Critical: all three drivers" with a churn rate of 100.0%, does not have the highest revenue at risk, which is actually 9225.97. A plausible business explanation is that a low rate applied to a very large base, such as the "Baseline: no drivers" segment with 2605 customers, can still produce substantial absolute loss, with revenue at risk of 8737.56. This suggests that targeting only the highest-rate segments would leave the largest dollar exposure unaddressed. To address this, the company should prioritize reviewing the "Baseline: no drivers" segment, with 2605 customers and revenue at risk of 8737.56, to identify potential retention strategies.

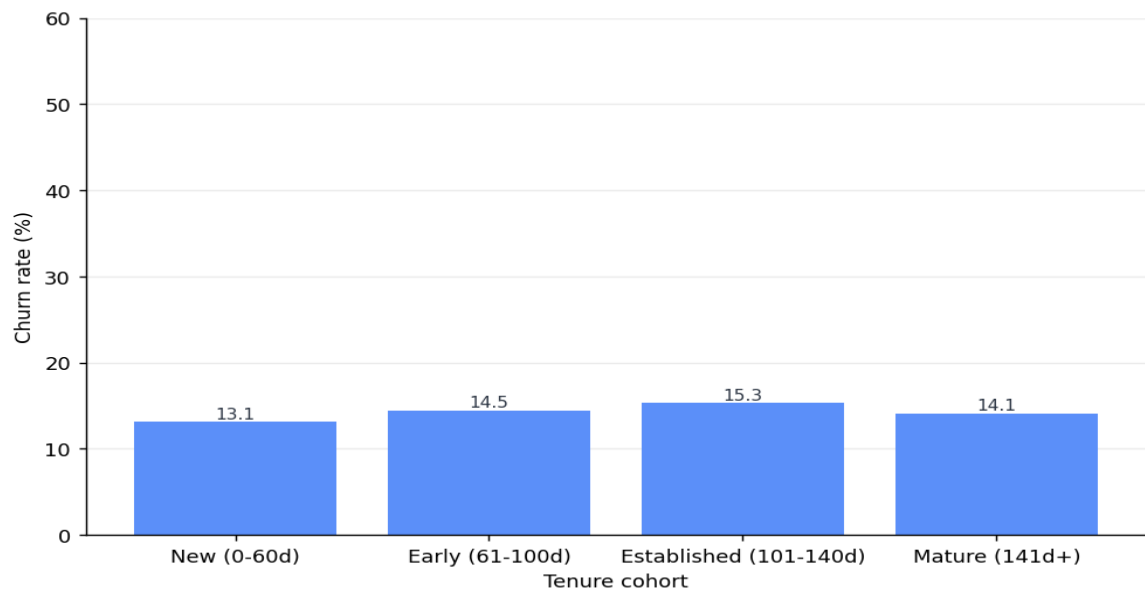
Tenure: an expected driver that is not one

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Cohort	Customers	Churn	ARPU	Avg service calls
New (0-60d)	511	13.11%	\$59.70	1.52
Early (61-100d)	1,153	14.48%	\$59.13	1.57
Established (101-140d)	1,138	15.29%	\$59.49	1.6
Mature (141d+)	531	14.12%	\$59.83	1.5

Portfolio structure by tenure cohort

Churn is flat across cohorts: 13.1% to 15.3%.

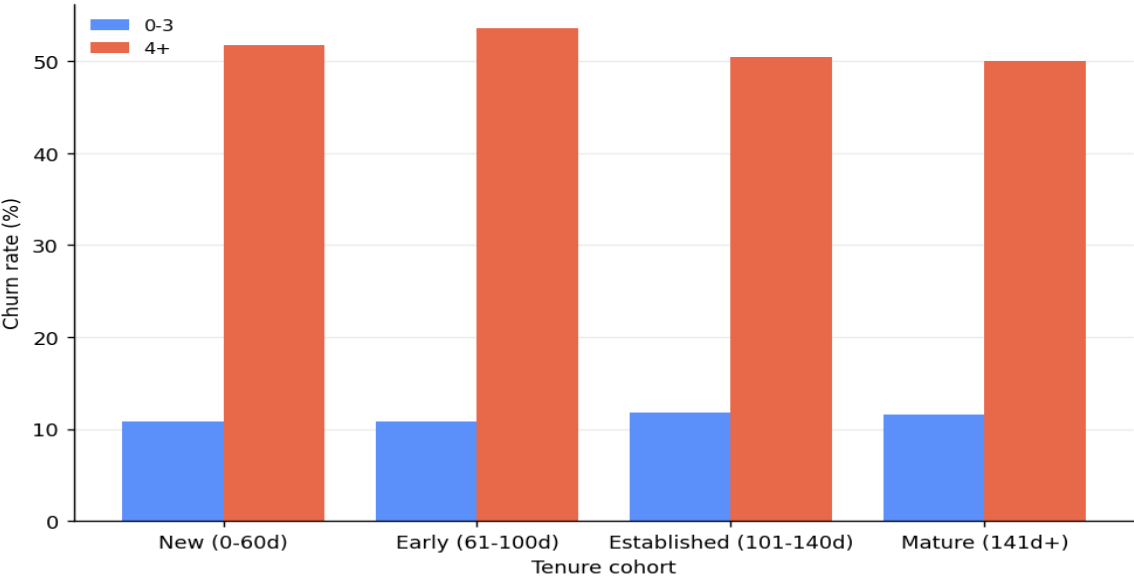


Plotted on the same y-scale as the driver charts to show how little variation there is here by comparison.

AI explanation. The pattern of churn rates across different tenure cohorts is relatively flat, ranging from 13.1% to 15.3%. This suggests that tenure does not predict churn, and targeting based on tenure may not be effective. The small spread between cohorts, with the highest churn rate of 15.29% in the Established cohort and the lowest of 13.11% in the New cohort, is not a meaningful pattern. To address the high churn rate, a specific step could be to review the service experience of customers who have made 4 or more service calls, as this may be a driver of churn, and consider implementing measures to improve their experience and reduce churn.

Churn by tenure cohort and service-call bucket

The service-call cliff holds in every cohort. Tenure itself does not predict churn.



Tenure is the ordered axis in this project because the dataset has no date column. It is useful for structure, not as a churn driver.

Portfolio structure over time‡

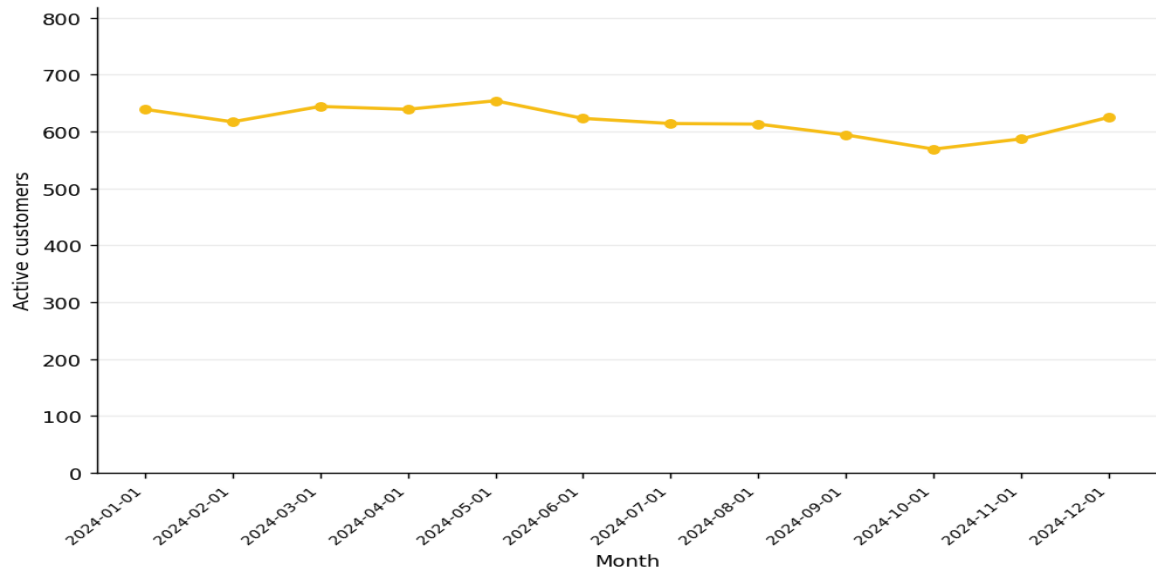
SIMULATED HISTORY. This section is derived from generated monthly snapshots, not observed history. It shows portfolio structure only. Churn in this panel is flat by construction and cannot support a trend claim or a forecast.

12	569-654	\$8,529-\$9,954
Months shown	Active base	Monthly revenue

Narration unavailable: Groq rate limit after 5 attempts. Lower concurrency or raise LLM_MIN_INTERVAL. {"error":{"message":"Rate limit reached for model `llama-3.3-70b-versatile` in organization `org\_01kyrzz9df89bszh3c4agqa5` service tier `on\_demand` on tokens per day (TPD): Limit 100000, Used 99026, Requested 2741. Please try again in 25m26.688s. Need more tokens? Upgrade to Dev Tier today at [https://console.groq.com/settings/billing](\"https://console.groq.com/settings/billing\")\""},\"type\":\"tokens\", \"code\":\"rate_limit_exceeded\"}}

Active customers by month (simulated history)

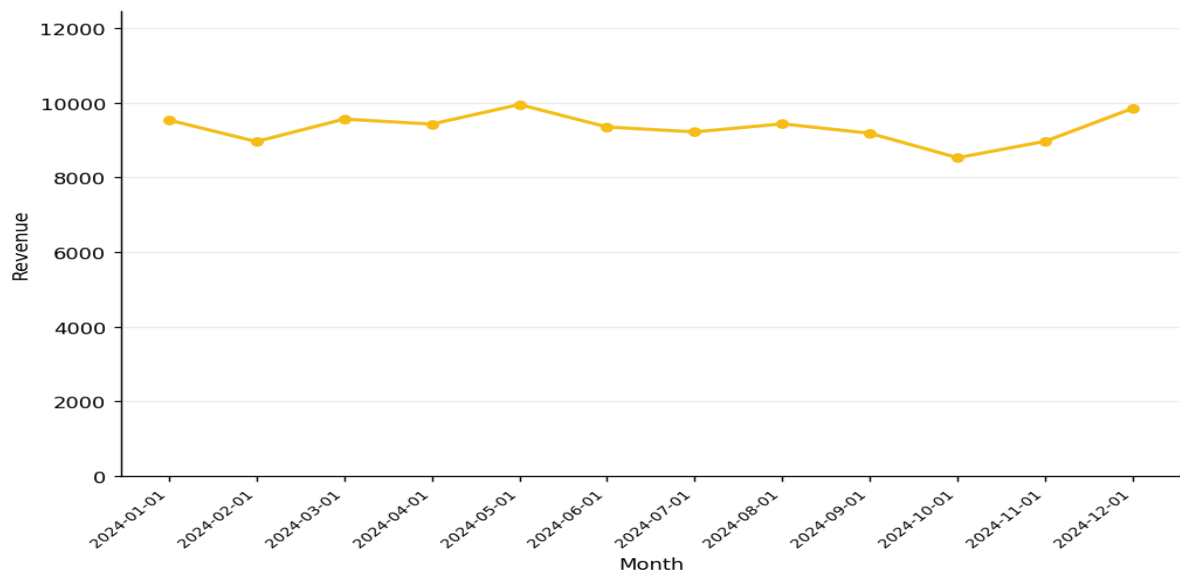
SIMULATED. Shows portfolio shape only — churn in this panel is flat by construction.



Generated from the snapshot, not observed. Included to show portfolio shape; it contains no information the snapshot did not already hold.

Monthly revenue (simulated history)

SIMULATED. Revenue derived by splitting each customer's real total across their active months.



Each customer's real total charge was distributed across their active months. Monthly totals sum back to the real portfolio revenue.

What this report cannot tell you

- **NO TIME DIMENSION.** The source data is a single snapshot with no dates. Nothing here describes how any metric changed over time, and no figure in this report is a forecast.
- **SIMULATED HISTORY SHOWS STRUCTURE, NOT TRENDS.** The monthly panel in this report is generated from the snapshot, so it contains no information the snapshot did not already hold. Churn in it is deliberately flat: a trend there would reflect the generator's random seed, not customer behaviour. It is included to show portfolio shape and tenure lifecycle, both of which are legitimate uses of derived monthly data.

- **TENURE IS NOT A DRIVER.** Churn is flat across tenure cohorts (13.1%-15.3%). This was expected to matter and does not - a genuine finding, and a reason not to target by tenure.
- **STATE-LEVEL FIGURES ARE NOISY.** 51 states average about 65 customers each, so a handful of churners moves a state several percentage points. Geography is the weakest dimension here.
- **PROJECTED VALUES REST ON ASSUMPTIONS.** Save rate, contact cost and acquisition cost are industry-typical placeholders, not measurements: the dataset contains no cost data. A save rate of 30% is assumed, with a sensitivity band of 15%-45%.
- **REVENUE IS PERIOD CHARGES.** Not monthly recurring revenue and not annualised.
- **CHURN RISK IS NOT PREDICTED HERE.** Segments describe which customers resemble those who already churned. Scoring individual customers is a separate modelling problem.

Provenance

Every observed figure traces to the source data. Segment totals reconcile to the portfolio: customers match (3,333), revenue matches (\$198,146.03).

Warning: narration in these sections failed validation and may contain unverified figures: kpi, segments, cohorts, structure.